HEINZ NIXDORF INSTITUT
UNIVERSITÄT PADERBORN

How To Train Your Large Language Model

Semester-end Presentation

Yven de Buhr, Joel Dag, Sashreek Nayak Dhaimodkar,
Luke Friedrichs, Jamil Mounzer, Martin Schröder

Supervisor: Prof. Dr. Axel Ngonga

DICE

Data Science Group
Paderborn University

July 22, 2025

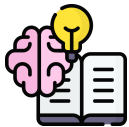
How To Train Your Large Language Model

Contents

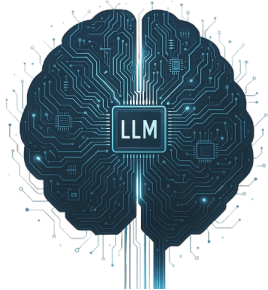
Introduction

Large Language Models

Rapid Knowledge Access



Flexibility



Productivity & Automation

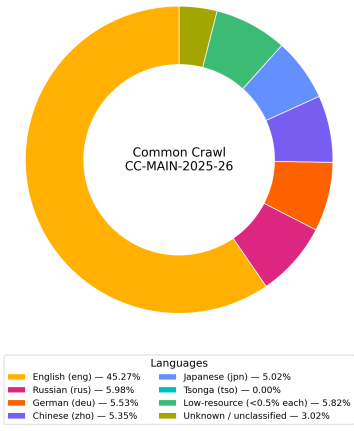
Multilingual Communication

Image sources: freepik.com, https://es.pinterest.com/pin/497436721353100015/

Challenges

Language Imbalance

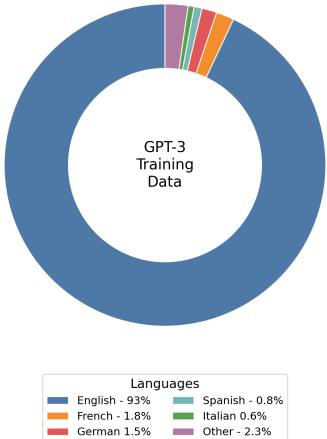
Common Crawl Language Distribution [1]



Challenges

Language Imbalance

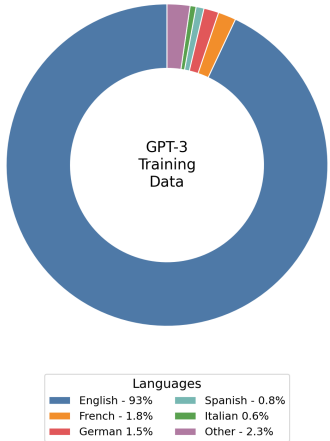
GPT-3 Training Data Language Distribution [2]



Challenges

Language Imbalance

GPT-3 Training Data Language Distribution [2]



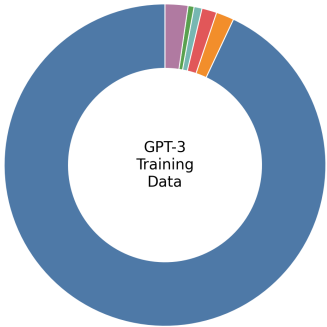
English-centric Bias

- Models strongly reflect **Anglophone worldview** [3]
- Marginalization of other cultures / norms [3]

Challenges

Language Imbalance

GPT-3 Training Data Language Distribution [2]



English-centric Bias

- Models strongly reflect **Anglophone worldview** [3]
- Marginalization of other cultures / norms [3]

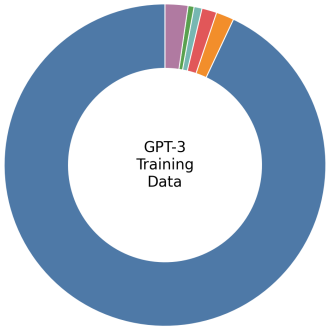
Performance Disparity

- LLMs show systematically **worse performance** on tasks in low-resource languages [4]

Challenges

Language Imbalance

GPT-3 Training Data Language Distribution [2]



English-centric Bias

- Models strongly reflect **Anglophone worldview** [3]
- Marginalization of other cultures / norms [3]

Performance Disparity

- LLMs show systematically **worse performance** on tasks in low-resource languages [4]

Technological Exclusion

- 2.191 / 2.485 tracked languages (**88.38%**) have zero known data in official databases [5]

Challenges

Curse of multilinguality

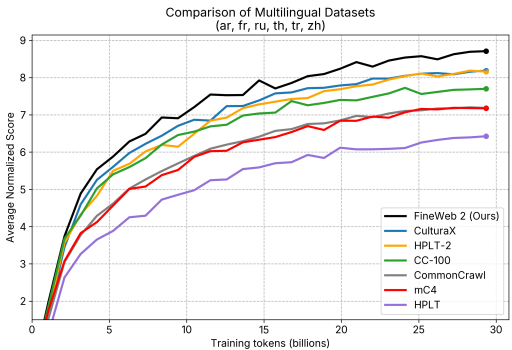
"As dataset sizes increase, adding multilingual data begins to **hurt performance** for **both** low-resource and high-resource languages, likely due to limited model capacity (the *“curse of multilinguality”*)". [6]

Objective of Project Group

Task: Train a large and **open-source multilingual** language model and address the challenges posed by the *curse of multilinguality*.

- Support 500+ languages
- Ensure computational efficiency
- Enable multimodal capabilities
- Maintain linguistic extensibility

Multilingual Dataset



Fineweb 2 Dataset [7]

- 20TB data
- 1800 languages
- 5 billion documents

Our subsample

- 14,000,000 documents in 190 languages

Workflow, Tools & Infrastructure Overview

Workflow & Tools

1. **Element Messaging Platform** (element.cs.uni-paderborn.de)
   - Communication and collaboration
2. **Kanban Board** (kanboard.cs.uni-paderborn.de)
   - Task visualization, progress tracking, and project workflow management
3. **Sciebo Cloud Storage** (uni-paderborn.sciebo.de)
   - Secure sharing of files and documents

Technology & Infrastructure

Permanent access

- **Noctua 2** Cluster (128× NVIDIA A100 40 GB) [8]
- **LOLA** GPU VM (2× NVIDIA H100-80GB)

Limited access

- **enexa2** GPU VM (2× *NVIDIA A100 80GB*)
- **morel** GPU VM (4× *NVIDIA H100 96GB*)
- **otus** GPU VM (4× *NVIDIA H100-94GB*)

Section 1 - Pretraining Approaches

Joint Multilingual Pretraining

Joint Multilingual Pretraining

- *From-scratch* implementation of the `GPT-2` architecture [9]
- 340M *parameters*
- Comparable *baseline* reference model

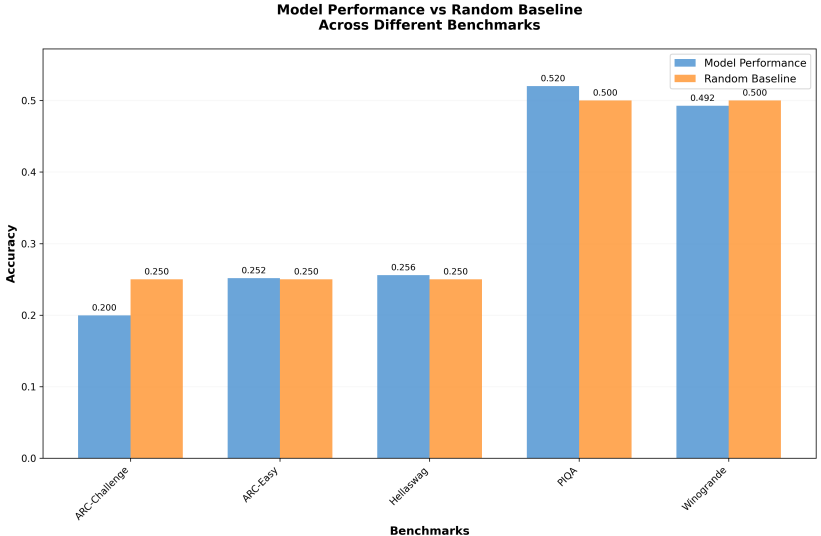
Joint Multilingual Pretraining

- *From-scratch* implementation of the `GPT-2` architecture [9]
- 340M *parameters*
- Comparable *baseline* reference model

Challenges:

- Very resource intensive
- *Curse of multilinguality*

Evaluation


Model Performance vs Random Baseline
Across Different Benchmarks
Model Performance
Random Baseline
Accuracy
0.0
0.1
0.2
0.3
0.4
0.5
0.200
0.250
0.252
0.250
0.256
0.250
0.520
0.500
0.492
0.500
ARC-Challenge
ARC-Easy
Hellaswag
PIQA
Winogrande
Benchmarks

Section 1 - Pretraining Approaches

Mixture of Experts (MoE)

Mixture of Experts (MoE)

- **Key Idea:** Only a few specialized experts are activated per input

Mixture of Experts (MoE)

- **Key Idea:** Only a few specialized experts are activated per input

Components:

- **Experts:** Specialized subnetworks trained to handle different data patterns
- **Router:** Chooses the most relevant experts dynamically

Mixture of Experts (MoE)

- **Key Idea:** Only a few specialized experts are activated per input

Components:

- **Experts:** Specialized subnetworks trained to handle different data patterns
- **Router:** Chooses the most relevant experts dynamically

Benefits:

- **Efficient** use of compute power
- **Scalable** model capacity
- Independent subnetworks specialized in different areas

Mixture of Experts (MoE)

- **Key Idea:** Only a few specialized experts are activated per input

Components:

- **Experts:** Specialized subnetworks trained to handle different data patterns
- **Router:** Chooses the most relevant experts dynamically

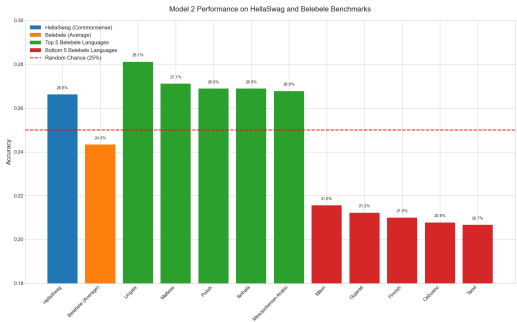
Benefits:

- **Efficient** use of compute power
- **Scalable** model capacity
- Independent subnetworks specialized in different areas

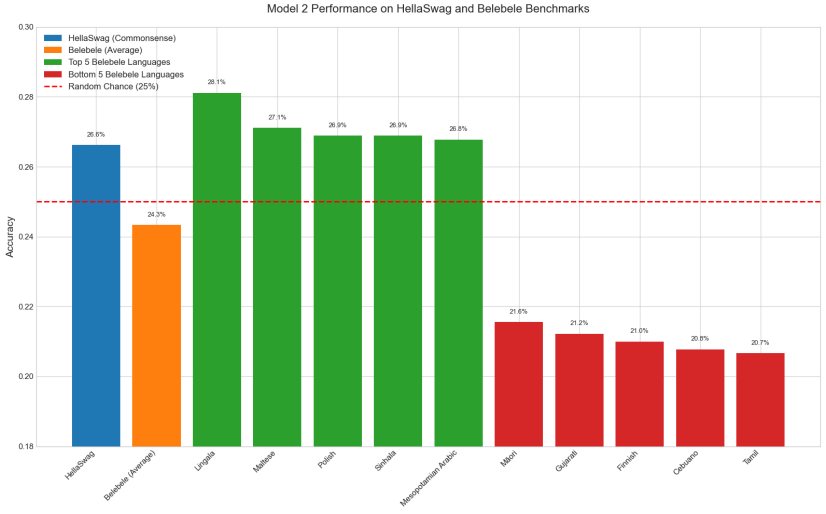
- **Multilingual Advantage:** Mitigation of the *curse of multilinguality*

MoE — Training Run #1

- `MIXTRAL` architecture [10]
- **Own Tokenizer:**
  14,850,045,952 *tokens*
- **Model Size:**
  - 203,727,360 **total** *parameters*
  - 47,227,776 **active** *parameters*
- `vocab_size` = 131,072
- 2 experts per token (`n_experts` = 7)
- 3 training epochs

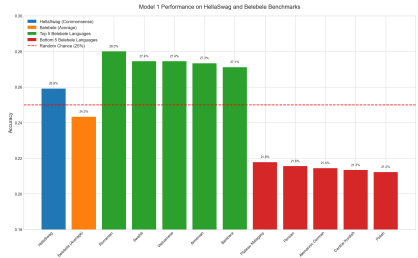


MoE - Training Run #1

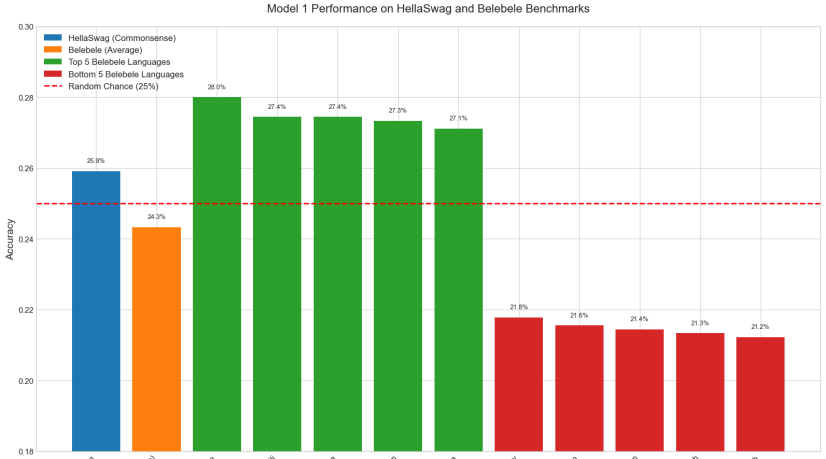


MoE — Training Run #2

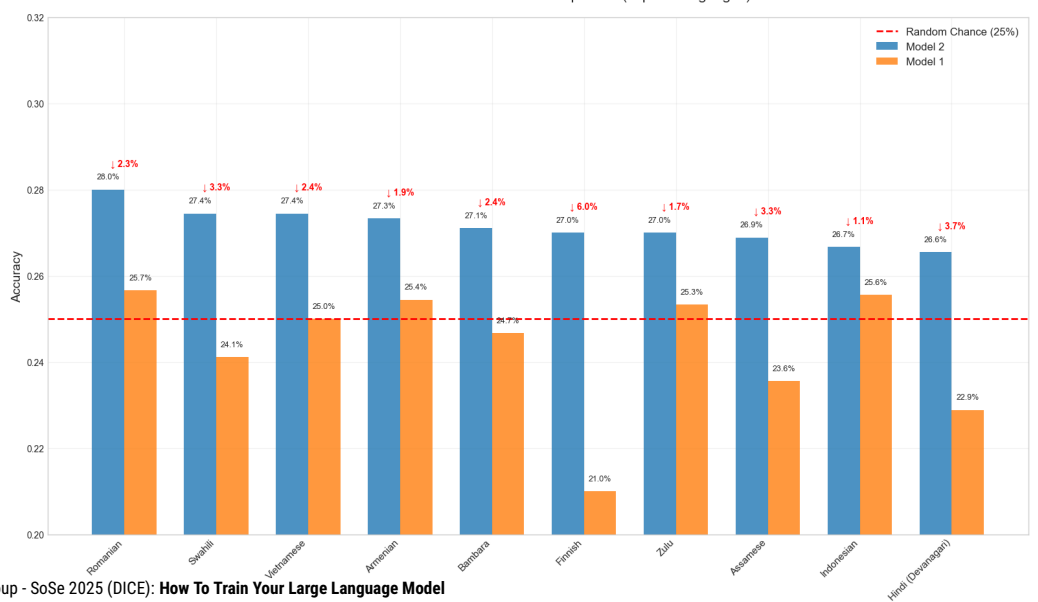
- `MIXTRAL` architecture [10]
- **Own Tokenizer:**
  22,105,942,016 *tokens* (⇑ *+ 7.25B*)
  (*byte fallback* enabled)
- **Model Size:**
  - 354,505,216 **total** *parameters* (⇑ *+ 150M*)
  - 177,558,016 **active** *parameters* (⇑ *+ 130M*)
- `vocab_size` = 131,072
- 2 experts per token (`n_experts` = 12 (⇑ *+ 5*))
- 1 training epoch (⇓ *- 2*)



MoE - Training Run #2



Comparison of Run #1 and Run #2



Summary of Run #2 - TFLOPs Profiler

- **Headline Metrics**
  - Overall GPU Utilization: **44.06** TFLOPS (*end-to-end* on **NVIDIA H100 80GB**)
  - Training Throughput: **89.2 samples/s** (2 GPUs)

Summary of Run #2 - TFLOPs Profiler

- **Headline Metrics**
  - Overall GPU Utilization: **44.06** TFLOPS (*end-to-end* on **NVIDIA H100 80GB**)
  - Training Throughput: **89.2 samples/s** (2 GPUs)
- **Latency per Step: 717 ms**
  - *Forward Pass* — 24% (175 ms) | 60 TFLOPS
  - *Backward Pass* — 61% (438 ms) | 48 TFLOPS *(primary bottleneck)*
  - *Optimizer Step* — 15% (104 ms)

Summary of Run #2 - TFLOPs Profiler

- **Headline Metrics**
  - Overall GPU Utilization: **44.06** TFLOPS (*end-to-end* on **NVIDIA H100 80GB**)
  - Training Throughput: **89.2 samples/s** (2 GPUs)
- **Latency per Step: 717 ms**
  - *Forward Pass* — 24% (175 ms) | 60 TFLOPS
  - *Backward Pass* — 61% (438 ms) | 48 TFLOPS *(primary bottleneck)*
  - *Optimizer Step* — 15% (104 ms)

Critical Bottleneck: MoE Load Imbalance

- Gating network overloads a few experts, leaving others idle.
- **Example** (Layer 0): *Most-used* expert = 52 GFLOPs vs. *least-used* = 5 GFLOPs
- **Result**: Busiest expert waiting, waste of compute cycles

Section 1 - Pretraining Approaches

Follow-up Steps

MoE Megatron

Megatron-LM

- GitHub: *https://github.com/NVIDIA/Megatron-LM*
- Useful for **scaling** models
- Enables *pipeline-* and *tensor parallelism*

MoE Megatron

Megatron-LM

- GitHub: *https://github.com/NVIDIA/Megatron-LM*
- Useful for **scaling** models
- Enables *pipeline-* and *tensor parallelism*

Our application

- *n* = 8 experts, `bf16` format, `expert_parallelism` = 4 (on **NVIDIA A100 40 GB**)
- **Median**: 214.1 TFLOP/s/GPU (**Min**: 22.7, **Max**: 225.7)
  (For reference: Overall GPU Utilization (MoE Run #2): 44.06 TFLOPS)

Section 2 - Finetuning Approaches

Cross-Lingual Finetuning

Cross-Lingual Transfer Learning

- **Idea:** Fine-tune existing model with multilingual data to demonstrate cross-lingual transfer effects, where performance improves for both trained languages and linguistically similar languages.

Cross-Lingual Transfer Learning

- **Idea:** Fine-tune existing model with multilingual data to demonstrate cross-lingual transfer effects, where performance improves for both trained languages and linguistically similar languages.

Challenges:

- Resource intensive: Fine-tuning requires substantial training relative to model size.
- *Curse of multilinguality:* Performance drops for unrelated languages not in training data.

Training setup + results

- Model: google/gemma-3-4b-pt
- Tokens: 113,191,379

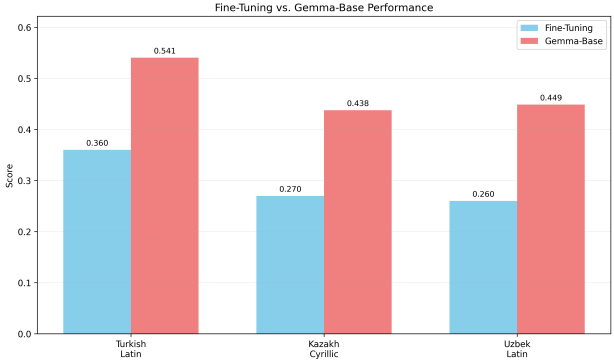


Figure: Fine-tuning on Turkish vs Gemma Base

Training setup + results

- Model: mistralai/Mistral-7B-v0.3
- Tokens: 263,916,337

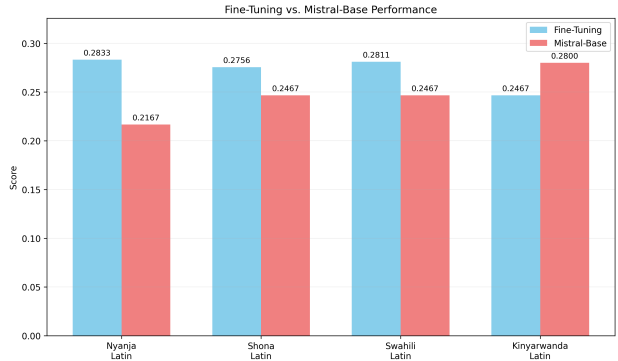


Figure: Fine-tuning on Swahili, Shona, and Nyanja vs Mistral Base

Section 2 - Finetuning Approaches

Adapters

Low-Rank Adaptation of Large Language Models (LoRA)

- Goal: Efficient fine-tuning for large models

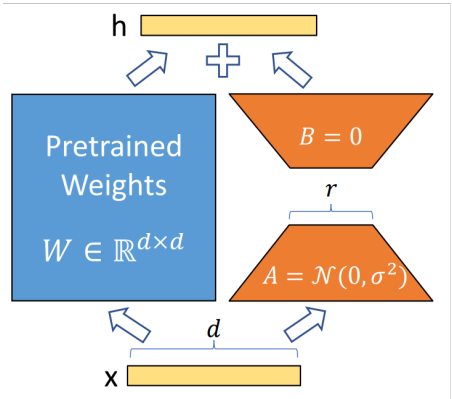


Figure: LoRA: Injecting low-rank updates into frozen pretrained weights [11]

Low-Rank Adaptation of Large Language Models (LoRA)

- Goal: Efficient fine-tuning for large models
- Freeze original weights $W \in \mathbb{R}^{d \times d}$

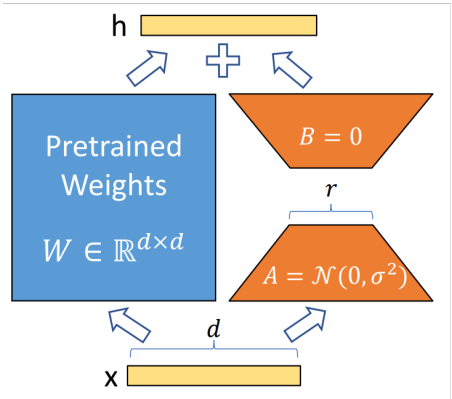


Figure: LoRA: Injecting low-rank updates into frozen pretrained weights [11]

Low-Rank Adaptation of Large Language Models (LoRA)

- Goal: Efficient fine-tuning for large models
- Freeze original weights $W \in \mathbb{R}^{d \times d}$
- Add trainable low-rank update: $\Delta W = BA$

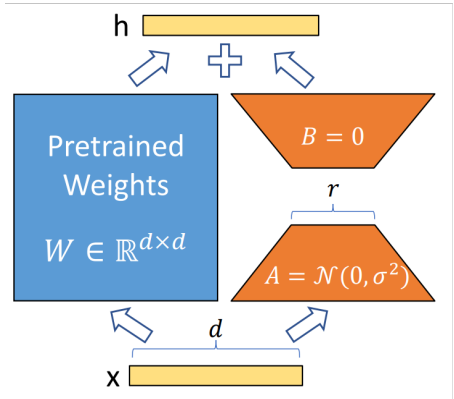


Figure: LoRA: Injecting low-rank updates into frozen pretrained weights [11]

Low-Rank Adaptation of Large Language Models (LoRA)

- Goal: Efficient fine-tuning for large models
- Freeze original weights $W \in \mathbb{R}^{d \times d}$
- Add trainable low-rank update: $\Delta W = BA$
  - $A \in \mathbb{R}^{r \times k}, \quad B \in \mathbb{R}^{d \times r}$
  - $r \ll \min(d, k)$

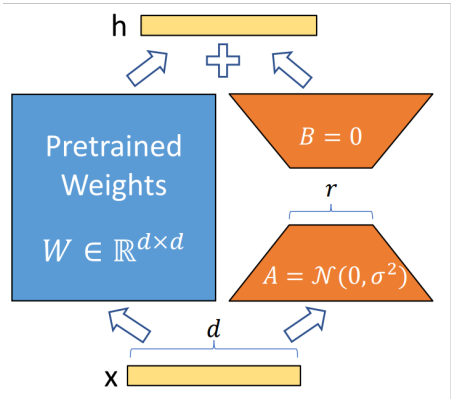


Figure: LoRA: Injecting low-rank updates into frozen pretrained weights [11]

Low-Rank Adaptation of Large Language Models (LoRA)

- Goal: Efficient fine-tuning for large models
- Freeze original weights $W \in \mathbb{R}^{d \times d}$
- Add trainable low-rank update: $\Delta W = BA$
  - $A \in \mathbb{R}^{r \times k}, \quad B \in \mathbb{R}^{d \times r}$
  - $r \ll \min(d, k)$
- Modified projection: $W'x = Wx + BAx$

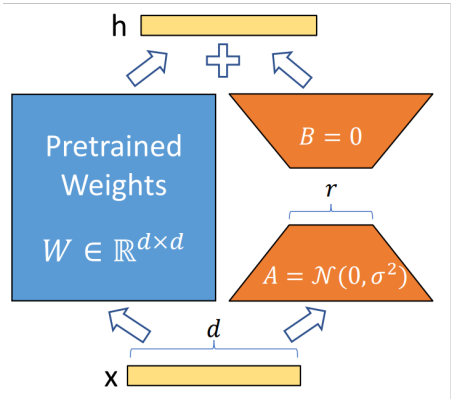


Figure: LoRA: Injecting low-rank updates into frozen pretrained weights [11]

Low-Rank Adaptation of Large Language Models (LoRA)

- Goal: Efficient fine-tuning for large models
- Freeze original weights $W \in \mathbb{R}^{d \times d}$
- Add trainable low-rank update: $\Delta W = BA$
  - $A \in \mathbb{R}^{r \times k}, \quad B \in \mathbb{R}^{d \times r}$
  - $r \ll \min(d, k)$
- Modified projection: $W'x = Wx + BAx$
- Only train A, B with significantly fewer parameters

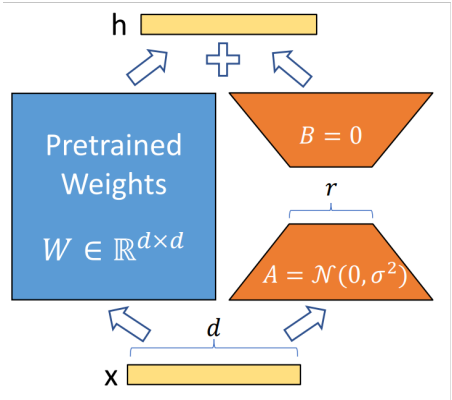


Figure: LoRA: Injecting low-rank updates into frozen pretrained weights [11]

Experimental Setup

Language Groups

- **Niger-Congo (Latin):** `swh, nso, zul, tsn, sot, yor, ibo, ewe, fon`
- **Arabic (Arabic):** `ars, apc, arb, acm, arz`
- **Cyrillic:** `kaz, khk`
- **Bengali (Bengali):** `asm, ben`
- **Georgian (Georgian):** `kat, xmf`

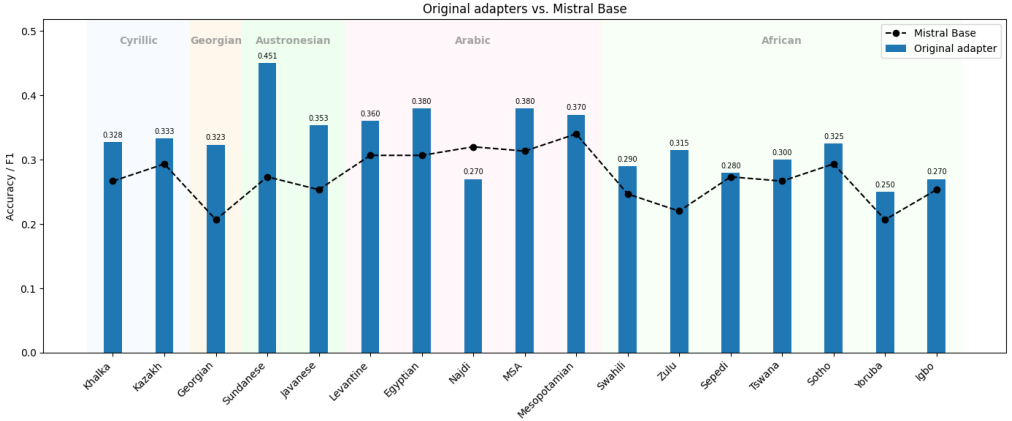
Model Setup

- Base: `Mistral-7B`
- Fine-tuned via: `LoRA/QLoRA adapters`
- Adapter size: **7.1M** parameters

Training Data (Tokens)

- Niger-Congo: **825M**
- Arabic: **602M**
- Bengali: **905M**
- Cyrillic: **205M**
- Georgian: **104M**

Evaluation and results



Direct Comparison: QLoRA Adapter vs Full Fine-Tuning vs Base

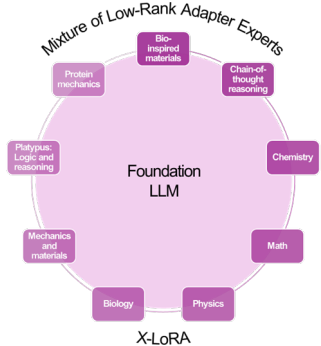
Table: Finetuning Mistral 7B on Swahili, Shona, and Nyanja reading comprehension (Belebele benchmark)

Model	Nyanja (nya)	Shona (sna)	Swahili (swh)	Total Score
Full Fine-Tuning	**0.2833**	**0.2756**	**0.2811**	**0.81**
QLoRA Adapter (Step 2505)	**0.2944**	**0.3422**	**0.3322**	**0.9689**
Mistral 7B Base	0.2467	0.2467	0.2467	0.7400

- Belebele reading comprehension tasks in low-resource languages.
- Improvement from near-random baseline to effective comprehension through fine-tuning.
- QLoRA captures much of the gain from full fine-tuning with far less compute.

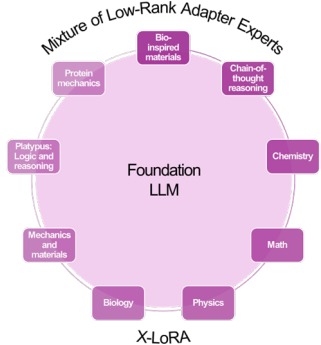
Mixture of Low-Rank Adapter Experts (XLoRA)

- **Goal:** Use a mixture of LoRA adapters dynamically during inference



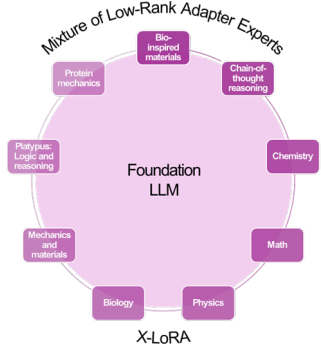
Mixture of Low-Rank Adapter Experts (XLoRA)

- **Goal:** Use a mixture of LoRA adapters dynamically during inference
- Freeze pretrained weights $W_0 \in \mathbb{R}^{d \times k}$



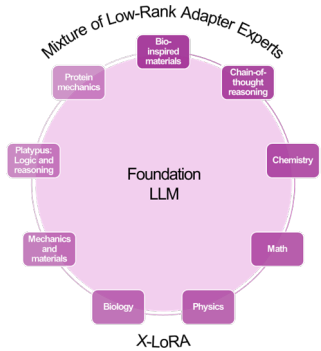
Mixture of Low-Rank Adapter Experts (XLoRA)

- **Goal:** Use a mixture of LoRA adapters dynamically during inference
- Freeze pretrained weights $W_0 \in \mathbb{R}^{d \times k}$
- Add multiple low-rank adapters: $\Delta W_i = B_i A_i$, for $i = 1, \ldots, N$



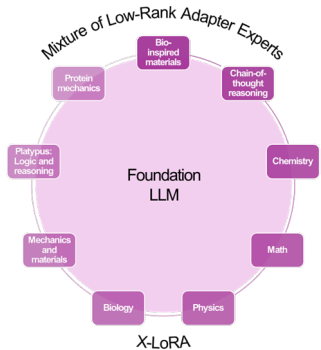
Mixture of Low-Rank Adapter Experts (XLoRA)

- **Goal:** Use a mixture of LoRA adapters dynamically during inference
- Freeze pretrained weights $W_0 \in \mathbb{R}^{d \times k}$
- Add multiple low-rank adapters: $\Delta W_i = B_i A_i$, for $i = 1, \ldots, N$
  - $B_i \in \mathbb{R}^{d \times r}, \quad A_i \in \mathbb{R}^{r \times k}$
  - $r \ll \min(d, k)$



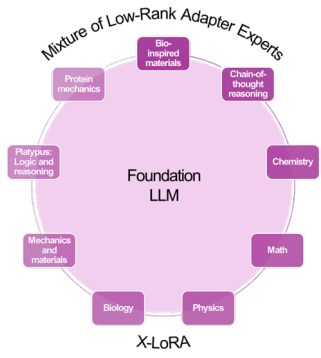
Mixture of Low-Rank Adapter Experts (XLoRA)

- **Goal:** Use a mixture of LoRA adapters dynamically during inference
- Freeze pretrained weights $W_0 \in \mathbb{R}^{d \times k}$
- Add multiple low-rank adapters: $\Delta W_i = B_i A_i$, for $i = 1, \ldots, N$
  - $B_i \in \mathbb{R}^{d \times r}, \quad A_i \in \mathbb{R}^{r \times k}$
  - $r \ll \min(d, k)$
- Forward pass becomes: $h = W_0 x + \Delta W_0 x = W_0 x + BAx$



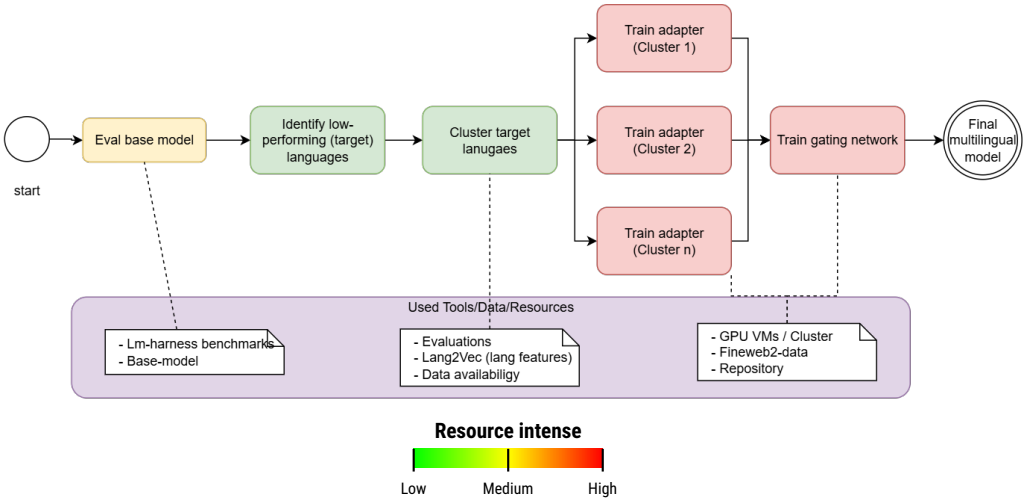
Mixture of Low-Rank Adapter Experts (XLoRA)

- **Goal:** Use a mixture of LoRA adapters dynamically during inference
- Freeze pretrained weights $W_0 \in \mathbb{R}^{d \times k}$
- Add multiple low-rank adapters: $\Delta W_i = B_i A_i$, for $i = 1, \ldots, N$
  - $B_i \in \mathbb{R}^{d \times r}, \quad A_i \in \mathbb{R}^{r \times k}$
  - $r \ll \min(d, k)$
- Forward pass becomes:
  $h = W_0 x + \Delta W_0 x = W_0 x + BAx$
- Each adapter's weight is scaled per token and layer: $\alpha_i^* = \alpha_i \cdot \lambda_i$
  where λ_i is predicted by a learned scaling head based on hidden states.

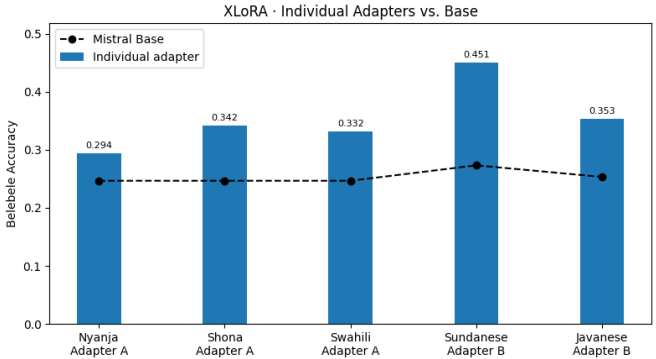


Mixture of Low-Rank Adapter Experts

Workflow of multilingulizing a base model

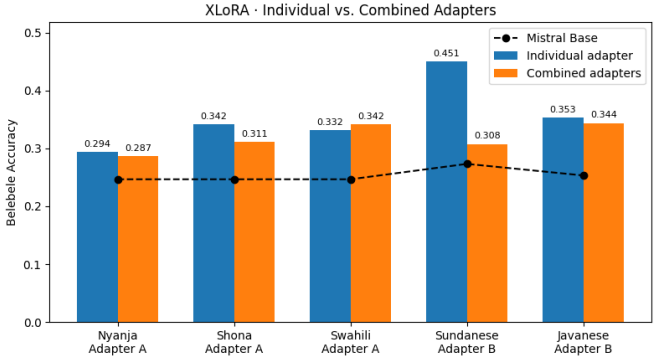


XLoRA Results: Individual vs. Combined Adapters



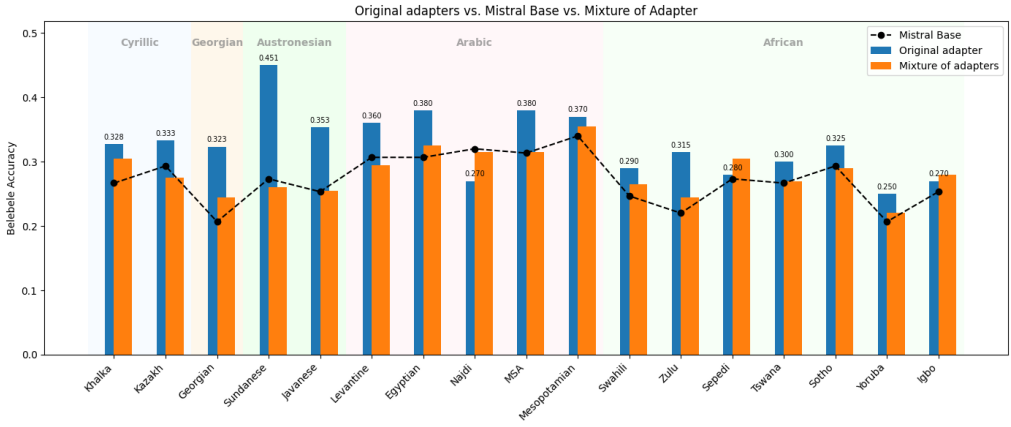
- Belebele reading comprehension benchmark

XLoRA Results: Individual vs. Combined Adapters



- Belebele reading comprehension benchmark
- Individual adapters perform stronger
- Gating decreases performance

XLoRA: Current Problems



- When scaling up the amount of languages, the Gating degrades most of the adapter improvements
- Work in Progress: More fine scaled gating implementation

Section 4

Current Problems / Future Work

Current Problems / Future Work

- **Tokenizer strategy**: single big tokenizer/ multiple individual ones?
- Only employ gating if *high confidence*, otherwise use base model.
- Combination of language adapters during inference?
- Figure out **optimal language clusters** for adapters.
- Counteract/ Mitigate **MoE Load Imbalance**.

Summary

Four approaches to explore

- Joint Multilingual Pretraining
- Mixture of Experts
- Cross-Lingual Transfer Learning
- Language Adapters

Three main challenges

- Curse of multilinguality
- Training efficiency
- Low-resource languages

Future Work

- **Tokenizer strategy**: single big tokenizer/ multiple individual ones?
- Only employ gating if *high confidence*, otherwise use base model.
- Combination of language adapters during inference?
- Figure out **optimal language clusters** for adapters.
- Counteract/ Mitigate **MoE Load Imbalance**.

Thank you!

Questions?

HTYLLM Project Group (SoSe 2025) at Paderborn University

Code: github.com/dice-group/HTYLLM-PG

References I

[1] "Statistics of Common Crawl Monthly Archives by commoncrawl — commoncrawl.github.io." https://commoncrawl.github.io/cc-crawl-statistics/plots/languages. [Accessed 20-07-2025].

[2] R. L. Johnson, G. Pistilli, N. Menédez-González, L. D. D. Duran, E. Panai, J. Kalpokiene, and D. J. Bertulfo, "The ghost in the machine has an american accent: value conflict in gpt-3," *arXiv preprint arXiv:2203.07785*, 2022.

[3] E. M. Bender, T. Gebru, A. McMillan-Major, and S. Shmitchell, "On the dangers of stochastic parrots: Can language models be too big?," in *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*, pp. 610–623, 2021.

[4] D. Blasi, A. Anastasopoulos, and G. Neubig, "Systematic inequalities in language technology performance across the world's languages," *arXiv preprint arXiv:2110.06733*, 2021.

[5] P. Joshi, S. Santy, A. Budhiraja, K. Bali, and M. Choudhury, "The state and fate of linguistic diversity and inclusion in the nlp world," *arXiv preprint arXiv:2004.09095*, 2020.

[6] T. A. Chang, C. Arnett, Z. Tu, and B. K. Bergen, "When is multilinguality a curse? language modeling for 250 high- and low-resource languages," 2023.

[7] "HuggingFaceFW/fineweb-2 · Datasets at Hugging Face — huggingface.co." https://huggingface.co/datasets/HuggingFaceFW/fineweb-2. [Accessed 21-07-2025].

References II

[8] C. Bauer, T. Kenter, M. Lass, L. Mazur, M. Meyer, H. Nitsche, H. Riebler, R. Schade, M. Schwarz, N. Winnwa, *et al.*, "Noctua 2 supercomputer," *Journal of large-scale research facilities JLSRF*, vol. 9, no. 1, 2024.

[9] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever, *et al.*, "Language models are unsupervised multitask learners," *OpenAI blog*, vol. 1, no. 8, p. 9, 2019.

[10] A. Q. Jiang, A. Sablayrolles, A. Roux, A. Mensch, B. Savary, C. Bamford, D. S. Chaplot, D. d. l. Casas, E. B. Hanna, F. Bressand, *et al.*, "Mixtral of experts," *arXiv preprint arXiv:2401.04088*, 2024.

[11] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, *et al.*, "Lora: Low-rank adaptation of large language models.," *ICLR*, vol. 1, no. 2, p. 3, 2022.

[12] N. Srivastava, D. Kuchelev, T. M. Ngoli, K. Shetty, M. Röder, H. Zahera, D. Moussallem, and A.-C. N. Ngomo, "Lola – an open-source massively multilingual large language model," 2025.

[13] A. Chronopoulou, D. Stojanovski, and A. Fraser, "Language-family adapters for low-resource multilingual neural machine translation," *arXiv preprint arXiv:2209.15236*, 2022.

[14] J. Pfeiffer, I. Vulić, I. Gurevych, and S. Ruder, "Mad-x: An adapter-based framework for multi-task cross-lingual transfer," *arXiv preprint arXiv:2005.00052*, 2020.

[15] HuggingFace, "Huggingfacefw/fineweb."

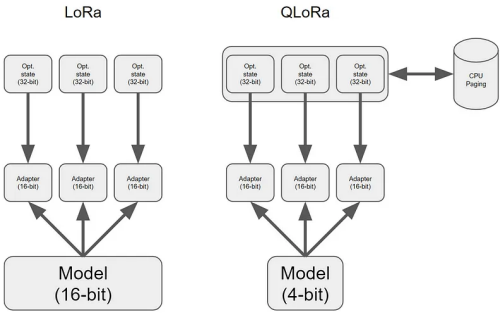
References III

[16] HuggingFace, “Huggingfacefw/fineweb-2.”

Section 3

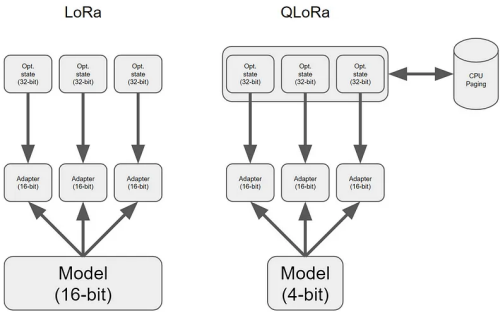
Backup Slides

QLoRA: Quantized LoRA



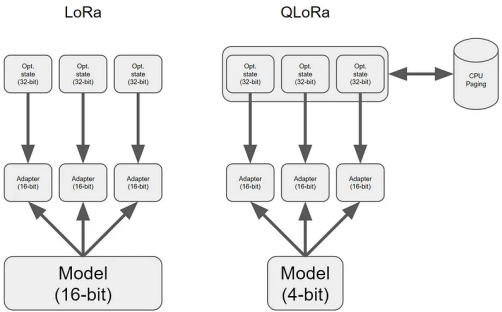
- TODO: remove and only include if we need more content
- Memory-efficient fine-tuning for large models

QLoRA: Quantized LoRA



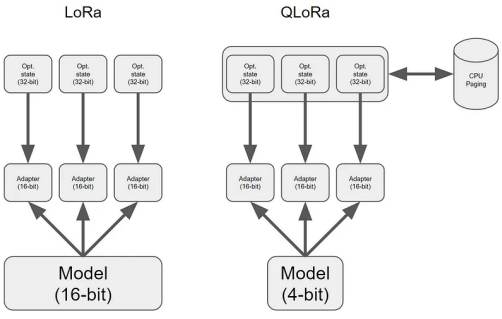
- TODO: remove and only include if we need more content
- Memory-efficient fine-tuning for large models
- **Quantize** base model to 4-bit

QLoRA: Quantized LoRA



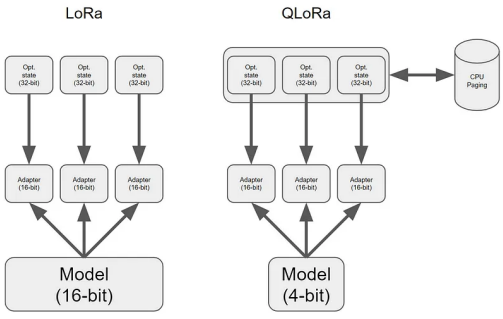
- TODO: remove and only include if we need more content
- Memory-efficient fine-tuning for large models
- **Quantize** base model to 4-bit
- **Freeze** quantized weights during training

QLoRA: Quantized LoRA



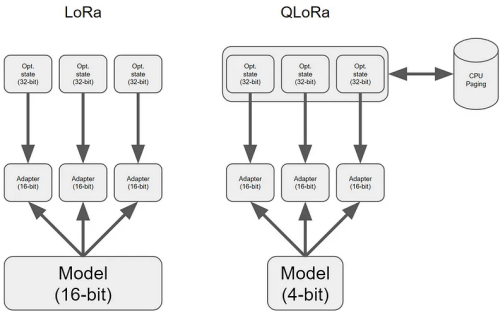
- TODO: remove and only include if we need more content
- Memory-efficient fine-tuning for large models
- **Quantize** base model to 4-bit
- **Freeze** quantized weights during training
- **Insert LoRA adapters:** $\Delta W = BA$

QLoRA: Quantized LoRA



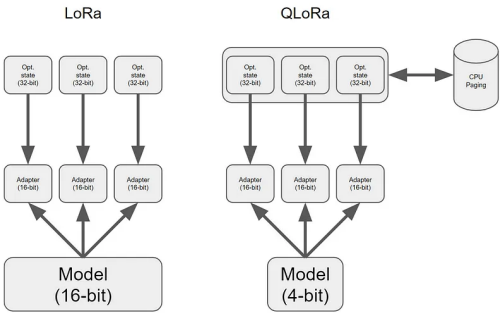
- TODO: remove and only include if we need more content
- Memory-efficient fine-tuning for large models
- **Quantize** base model to 4-bit
- **Freeze** quantized weights during training
- **Insert LoRA adapters:** $\Delta W = BA$
  - $A \in \mathbb{R}^{r \times k}, \quad B \in \mathbb{R}^{d \times r}$
  - $r \ll \min(d, k)$

QLoRA: Quantized LoRA



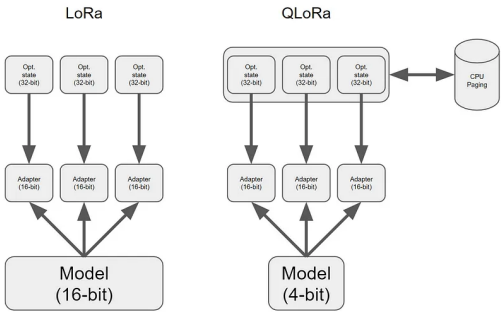
- TODO: remove and only include if we need more content
- Memory-efficient fine-tuning for large models
- **Quantize** base model to 4-bit
- **Freeze** quantized weights during training
- **Insert LoRA adapters:** $\Delta W = BA$
  - $A \in \mathbb{R}^{r \times k}, \quad B \in \mathbb{R}^{d \times r}$
  - $r \ll \min(d, k)$
- Forward pass: $W'x = Wx + BAx$

QLoRA: Quantized LoRA



- TODO: remove and only include if we need more content
- Memory-efficient fine-tuning for large models
- **Quantize** base model to 4-bit
- **Freeze** quantized weights during training
- **Insert LoRA adapters:** $\Delta W = BA$
  - $A \in \mathbb{R}^{r \times k}, \quad B \in \mathbb{R}^{d \times r}$
  - $r \ll \min(d, k)$
- Forward pass: $W'x = Wx + BAx$
- Only train A, B; rest is 4-bit frozen

QLoRA: Quantized LoRA



- TODO: remove and only include if we need more content
- Memory-efficient fine-tuning for large models
- **Quantize** base model to 4-bit
- **Freeze** quantized weights during training
- **Insert LoRA adapters:** $\Delta W = BA$
  - $A \in \mathbb{R}^{r \times k}, \quad B \in \mathbb{R}^{d \times r}$
  - $r \ll \min(d, k)$
- Forward pass: $W'x = Wx + BAx$
- Only train A, B; rest is 4-bit frozen
- Enables fine-tuning models more compute efficiently